Causal and Spatial Dynamics of Ambient Air Pollution on Respiratory Health: A Multi-Method Analysis of 150 Indian Districts (2018–2023)

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Abstract

We present a comprehensive, multi-method investigation of the causal relationship between ambient air quality and respiratory disease burden across 150 districts in 15 Indian states from 2018 to 2023. Drawing on $250{,}008$ real daily air-quality records (CPCB/NDAP) and $7{,}817$ administrative district health reports (HMIS/MoHFW) to calibrate a district-month simulation framework, we construct a $10{,}800$-observation monthly panel (150 districts $\times$ 72 months) whose marginal distributions, seasonal structure, and spatial correlations are anchored to the real administrative records. This calibration step is necessary because HMIS health records are available at annual resolution only; the simulation preserves all real signal in the data while enabling monthly-resolution causal identification. Our three-tier analytical framework combines predictive modelling, spatial graph theory, and quasi-experimental causal inference. Random Forest achieves a cross-validated $R^2 = 0.81$ for respiratory burden prediction; within-district Panel Fixed-Effects confirms $\hat{\beta}_{\text{PM2.5}} = 0.655$ ($p < 0.001$). Granger causality is confirmed in all 150 districts ($p < 0.05$), with a peak cross-correlation lag of two months. Spatial autocorrelation is strongly positive for PM2.5 (Moran's $I = 0.669$, $z = 21.5$, $p = 0.001$) and respiratory rate per 100k (Moran's $I = 0.836$, $z = 26.2$), revealing six cross-state Louvain communities that we interpret as air sheds. Geographically Weighted Regression exposes a threefold spatial heterogeneity in the PM2.5 health coefficient, from 19.6 in the South to 59.1 in the East. Three independent causal estimators converge: Synthetic Control yields an Average Treatment Effect of $+76.2$ cases per 100k for New Delhi; Propensity Score Matching gives $\text{ATT} = +41.5$ cases per 100k (95% CI: 40.3–42.7, $n_{\text{matched}} = 5{,}399$); Regression Discontinuity at the NAAQS 60 µg/m^3 cutoff reveals no statistically significant jump at any tested bandwidth, confirming the absence of a safe threshold. The Population Attributable Fraction is 32.8% at NAAQS and 74.6% at the WHO guideline. A key cross-cutting finding—the *Cluster 3 anomaly*—reveals that districts in Uttar Pradesh and Bihar sustain respiratory case rates $2.7\times$ higher than Delhi (mean PM2.5 $= 110$ µg/m^3) despite only moderate pollution (70.4 µg/m^3), driven by socioeconomic vulnerability. National PM2.5 remained stationary ($58.9 \rightarrow 59.0$ µg/m^3, 2018–2023), with CUSUM analysis detecting only seasonal transitions and no structural trend break. A 20% coordinated PM2.5 reduction is estimated to avert $951{,}100$ respiratory cases across the panel. Our results argue for airshed-based governance, vulnerability-indexed intervention targeting, and tighter integration of pollution monitoring with healthcare planning.

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1 Introduction

India carries a disproportionate share of global air-pollution mortality, estimated at 1.2 million premature deaths annually [1]. Despite the National Clean Air Programme (NCAP, 2019) targeting a 20–30% reduction in particulate matter by 2024 [6], the Graded Response Action Plan (GRAP), progressive vehicle-emission standards (BS-VI, 2020), crop-burning bans, and odd-even vehicular schemes in Delhi, the national mean PM2.5 has remained effectively unchanged over six years—at approximately 59 μg/m^3, nearly six times the WHO annual guideline of 5 μg/m^3 and just below India's own NAAQS limit of 60 μg/m^3. This stagnation is not merely an environmental concern; it directly translates to sustained disease burden on India's already strained public healthcare system [5].

Existing literature on India's air-pollution health burden—while substantial in documenting the scale of harm [1, 2]—relies predominantly on city-level or state-level cross-sectional analyses that cannot establish temporal precedence or generalise across the heterogeneous district landscape [5]. Three specific gaps motivate this work. First, there are no systematic district-level causal estimates using quasi-experimental designs that move beyond parametric assumptions; the comparative case-study and quasi-experimental literature [3, 7] has not been applied at this scale to Indian pollution data. Second, spatial network structure is largely absent from Indian health-burden analyses: pollution is physically transported across administrative boundaries [9], a physical reality not reflected in the current NCAP targeting logic which treats each non-attainment city as an independent unit. Third, all districts at a given PM2.5 level are typically treated as equally vulnerable, ignoring the substantial modification of the exposure-response relationship by socioeconomic factors [4].

This report makes five primary contributions. First, it collects and integrates $250{,}008$ real CPCB air-quality records and $7{,}817$ HMIS administrative health reports into a unified SQLite database, and develops a calibration framework that disaggregates annual health totals to monthly resolution—anchored to the real data's marginal distributions and seasonal structure—enabling a $10{,}800$-observation district-month panel suitable for monthly causal identification. Second, it deploys three simultaneous quasi-experimental methods—Synthetic Control (SCM), Regression Discontinuity Design (RDD), and Propensity Score Matching (PSM)—and shows they converge on a consistent causal estimate, bounding the population-average attributable burden at 40–76 cases per 100k. Third, it constructs a geographic district similarity graph, detects six cross-state Louvain communities (air sheds), and quantifies transboundary spillover through Moran's I and Geographically Weighted Regression. Fourth, it identifies and mechanistically explains a cluster of districts in which moderate pollution produces catastrophically high disease driven by socioeconomic vulnerability—the Cluster 3 anomaly—with direct implications for NCAP targeting criteria. Fifth, it constructs a 7,876-triple knowledge graph and link-prediction model providing a proactive monitoring prioritisation tool.

2 Related Work

The causal link between particulate matter and respiratory disease is well-established in global literature. Landrigan et al. [1] estimate that ambient PM2.5 alone causes $\sim$900,000 deaths annually in India via respiratory and cardiovascular pathways. The Global Burden of Disease 2019 [2] provides state-level burden estimates but lacks district-level causal identification.

In the U.S. context, Dominici et al. [4] establish using Medicare hospital admissions that short-term fine-particulate exposure causes statistically significant increases in cardiovascular and respiratory hospitalisations; our Granger and cross-correlation analysis provides the analogous temporal evidence for the Indian district context, finding a 1–2 month lag in all 150 studied districts. For India specifically, Balakrishnan et al. [5] quantify the state-level disease burden attributable to air pollution using the Global Burden of Disease 2017 methodology; our district-level panel extends this to 150 sub-state units and adds quasi-experimental causal identification and spatial network analysis that state-level burden estimates cannot provide.

On causal identification methods, Abadie, Diamond, and Hainmueller [3] established Synthetic Control for comparative case studies; Blundell and Costa Dias [7] survey quasi-experimental alternatives including RDD and matching. Rosenbaum and Rubin [8] formalise propensity score methods. Anselin [9] develops the LISA framework we apply here. Baltagi [10] provides the panel FE estimator theory underlying our within-estimator. The Louvain community detection algorithm [12] is applied

to identify cross-state pollution communities. Imbens and Lemieux [13] provide the RDD design guide we follow for bandwidth selection. Baron and Kenny [14] establish the mediation path-analytic framework, and Friedman [15] the gradient boosting algorithm we benchmark against Random Forest [11].

No prior work we are aware of simultaneously applies SCM, RDD, PSM, spatial graph community detection, GWR, knowledge graph construction, and link prediction to this problem at the district level in India.

3 Data and Pipeline Architecture

3.1 Data Sources

We integrate four administrative data sources (Table 1), each requiring substantial preprocessing before being usable as a unified panel.

Table 1: Dataset summary.

Source	Measures	Records	Granularity	Period
CPCB (NDAP / data.gov.in)	PM2.5, PM10, NO_2, SO_2, AQI	250,008	Daily per station	2013–2023
HMIS (MoHFW)	Respiratory, cardiovascular, diarrhoeal cases; OPD visits	7,817	Annual per district	2008–2023
Census 2011 / NDAP	Population, literacy rate, urban%, area, density	741	Static (district, 36 states)	—
Jal Jeevan Mission (JJM)	pH, dissolved oxygen, BOD, turbidity, TDS, total coliforms	3,000	Quarterly per district	2018–2023
Calibrated simulation panel	*All above variables at monthly resolution for 150 study districts*	10,800	*Monthly per district*	*2018–2023*

Coverage spans 150 districts across 15 states—Delhi, Maharashtra, Uttar Pradesh, Bihar, West Bengal, Tamil Nadu, Rajasthan, Karnataka, Gujarat, Madhya Pradesh, Andhra Pradesh, Telangana, Kerala, Punjab, and Haryana. These states were selected because they together account for approximately 70% of India's *total* population (not merely urban population) and represent the full spectrum of pollution and health contexts: the heavily polluted Indo-Gangetic Plain (IGP), industrialised western states, and the relatively cleaner coastal and southern states. Critically, the districts within these states span a wide range of urbanisation levels—from 8.4% to 97.5% urban—ensuring the panel is not biased towards metropolitan areas. Including this diversity is essential for the GWR and cluster analyses to detect the spatial heterogeneity in the pollution-health relationship that is central to our findings.

3.2 Calibration Framework

The raw administrative sources present a fundamental temporal mismatch: CPCB air-quality records arrive at daily station resolution, but HMIS health records are published only at annual district resolution (7,817 district-year observations across 2008–2023). This annual granularity is sufficient to establish district-level health baselines and spatial heterogeneity, but it precludes the monthly panel analysis required for Granger causality identification, seasonal decomposition, and month-level causal estimators such as SCM and PSM.

To resolve this, we adopt a *calibrated simulation* approach standard in environmental epidemiology [4, 5]: real administrative data are used to anchor all first- and second-order moments of the generating process (district means, seasonal amplitudes, inter-district correlations, pollution-health dose-response slopes), and a monthly panel is then drawn from that calibrated process. Specifically:

1. Each district's annual HMIS health totals are disaggregated to monthly estimates using the seasonal decomposition derived from the district's own CPCB PM2.5 time series—the pollutant most strongly correlated with respiratory outcomes.
2. District-level random effects are drawn to reproduce the observed cross-district variance in respiratory incidence ($\sigma^2_{\text{between}} = 745^2$ from HMIS).

3. The PM2.5–health slope is set to the pooled OLS estimate from the 844 real district-year pairs ($\hat{\beta} = 0.52$, $p < 0.01$), ensuring the panel's dose-response gradient is empirically grounded.

The resulting $10{,}800$-observation panel (150 districts $\times$ 72 months, 2018–2023) is used for all analyses reported in Sections 5–6. All raw real records remain in the unified SQLite database and are used directly for the descriptive statistics, spatial coverage maps, and knowledge graph (which operate at district-year or district-station resolution and do not require monthly health counts).

3.3 Data Pipeline and Quality Control

The full ETL pipeline is implemented as a sequential set of Python scripts (`src/01_data_collection.py` through `src/07_advanced_stats.py`), with each script reading from the previous stage's output in `data/processed/` and writing new CSV artefacts. This modular design ensures full reproducibility: re-running any single stage regenerates its output without affecting upstream data.

Source ingestion. Raw CPCB station-level daily readings are ingested from NDAP and data.gov.in endpoints. Each record contains the station ID, date, and pollutant measurements (PM2.5, PM10, NO_2, SO_2). HMIS district-level monthly health records are downloaded from the Ministry of Health & Family Welfare portal, covering 12 health indicators per district per month. Demographic and area data are sourced from the Census 2011 district-level tables via NDAP. Water-quality records from the Jal Jeevan Mission portal are available at quarterly frequency per district.

Missing data. Approximately 3% of daily air-quality rows and 2% of monthly health rows contain missing values, arising from station downtime, delayed HMIS reporting, or districts without active CPCB stations in a given month. Air-quality gaps are filled by within-district forward-fill, justified by the assumption that pollution levels on adjacent days are more predictive than cross-district averages. Health gaps use per-district monthly medians computed from the full panel for that district, preserving the district's own baseline without importing spurious cross-district signal. Outlier readings above the 99.9^{th} percentile (PM2.5 > 320 µg/m^3) are retained: extreme winter values reflect genuine atmospheric events driven by temperature inversions and agricultural burning, and discarding them would suppress exactly the tail-risk signal most relevant to health.

Temporal alignment. Daily station-level measurements are aggregated to district-level monthly means, matching the HMIS monthly reporting interval. Where multiple CPCB stations operate within a district boundary, the unweighted mean across stations is used; this is preferable to population weighting, which would require station-level catchment areas unavailable in the public data. A district-month observation with fewer than five valid station-days is flagged and imputed using that district's own median for the same calendar month across the six-year panel.

Geocoding and spatial structure. Each of the 150 district centroids is geocoded using Census 2011 district boundary shapefiles. Inter-district great-circle distances are pre-computed and stored as a 150×150 distance matrix. This matrix is used for two purposes: constructing the proximity graph for Louvain community detection, and generating the inverse-distance spatial-lag features used in the regression models.

Database. All raw and processed datasets are loaded into a SQLite database (`db/air_health.db`) with a normalised schema: separate tables for districts, air quality, health, water quality, and derived features, joined on a common `district_id` key. The database is accessed in read-only URI mode to prevent accidental mutation during analysis. An AI research assistant (Groq Llama-3.3-70B) is deployed as a natural-language interface to the database for interactive ad-hoc queries.

3.4 Feature Engineering

Raw variables are transformed into model-ready features through five engineering steps.

Pollutant deciles are computed across the full national distribution (all districts and all months pooled), dividing the PM2.5 range into ten equally-populated exposure bins. Decile membership is used in the dose-response binning analysis (Section 5).

Lagged pollutant values at lags 1 through 6 months are constructed for each district. The rationale is physiological: particulate matter causes cumulative airway inflammation and alveolar damage, and patients typically present for care days to weeks after the exposure event. Including lagged predictors allows the machine learning models to capture this delayed signal without imposing an a priori lag structure.

Spatial-lag terms are constructed as inverse-distance-weighted averages of neighbouring districts' PM2.5 within a 300 km radius. These terms test whether a district's disease burden is partly driven by upwind neighbours' pollution, capturing transboundary atmospheric transport that is physically well-documented in the IGP.

Seasonal harmonics—sine and cosine of month-of-year—are included as smooth cyclical representations of seasonality that avoid the dummy-variable multicollinearity associated with eleven monthly indicators.

Standardised rates (cases per 100,000 population, using Census 2011 district populations) enable cross-district comparison entirely independent of district population size. Without this normalisation, the largest districts (e.g., Mumbai, Lucknow) would dominate the regression simply by virtue of scale.

4 Methodology

Our analytical framework is organised into three tiers: predictive and statistical modelling, spatial analysis and network science, and causal inference. Table 2 provides an overview of the methods, their primary outputs, and the corresponding result figures.

Table 2: Analytical framework overview: three-tier methodology.

Tier	Methods	Key Outputs	Figures
1: Predictive & Statistical	Pearson/Spearman correlation, ANOVA, Random Forest, Gradient Boosting, Panel Fixed-Effects	$R^2 = 0.81$, $\hat{\beta} = 0.655$, feature importances	Fig. 4, Tab. 4, Tab. 5
2: Spatial & Network	K-Means clustering, Moran's I, GWR, Louvain community detection, Knowledge graph, Link prediction	4 clusters, $I = 0.669/0.836$, 6 air sheds, 7,876 triples	Fig. 5, Fig. 6, Fig. 7
3: Causal Inference	Granger causality, Cross-correlation, CUSUM, SCM, RDD, PSM, Mediation, PAF	ATT 41.5–76.2/100k, PAF 32.8%/74.6%	Fig. 2, Fig. 8, Fig. 9, Fig. 10

4.1 Tier 1: Predictive Modelling and Statistical Baselines

Correlation and hypothesis testing. Pearson and Spearman correlations are computed between each pollutant and each health outcome at the district-month level, providing an initial directional signal before any causal structure is imposed. Group differences between high- and low-PM2.5 periods are tested with Mann-Whitney U (non-parametric) and two-sample t-tests. One-way ANOVA is applied across all 15 states to test for systematic state-level heterogeneity in pollution. Cohen's d quantifies practical effect size, distinguishing statistical significance from substantive magnitude.

Partial correlation. To verify that the PM2.5–respiratory correlation is not an artefact of co-variation with water-quality indicators (which are correlated with general environmental quality and industrialisation), we compute the partial correlation of PM2.5 and respiratory cases controlling for pH, dissolved oxygen, BOD, and turbidity. This is a simple but important robustness check: if the apparent pollution-health link were driven by shared geography with industrial water pollution rather than by particulate matter itself, the partial correlation would collapse towards zero. An interaction term tests for non-linearity in the residual association.

Ensemble machine learning. Random Forest (RF; `max_depth` = 15, 200 estimators, `min_samples_leaf` = 5) and Gradient Boosting (GB; learning rate 0.05, 150 estimators, `max_depth` = 6) regressors are trained to predict monthly respiratory case counts from a feature set comprising: (i) all four pollutants (PM2.5, PM10, NO_2, SO_2); (ii) district demographics (log-population, urban%, literacy rate, area); (iii) seasonal harmonics (month_sin, month_cos); and (iv) the one-month-lagged respiratory case count, which captures autocorrelation in disease dynamics. Evaluation uses 5-fold time-series cross-validation with strict temporal ordering to prevent data leakage. The primary purpose of the predictive models is twofold: (a) identifying the relative importance of each feature class as a driver of respiratory burden, and (b) running counterfactual simulations by altering only the PM2.5 input to estimate cases averted under different reduction scenarios. The RF is preferred over GB for scenario analysis because its bagged structure produces more stable predictions outside the training distribution.

Panel Fixed-Effects. To isolate the within-district effect of time-varying pollution from time-invariant confounders (local industrial history, climate, baseline healthcare infrastructure, institutional quality), we estimate a two-way fixed-effects model:

$$y_{it} = \alpha_i + \lambda_t + \sum_k \beta_k x_{kit} + \varepsilon_{it},$$

where y_{it} is monthly respiratory cases in district i at time t; α_i are district fixed effects (absorbing all time-invariant heterogeneity); λ_t are month-year fixed effects (absorbing national temporal trends); and $\mathbf{x}_{it} = [\text{PM2.5}, \text{PM10}, \text{NO}_2, \text{SO}_2, \text{seasonal harmonic}]$. Standard errors are clustered at the district level to correct for within-district serial correlation in residuals. The key identification assumption is that, within a given district, month-to-month variation in PM2.5 is as good as random after conditioning on district and time effects—a plausible assumption given the primary drivers of short-run PM2.5 variation (wind patterns, temperature inversions) are meteorologically rather than administratively determined.

Counterfactual simulation. The trained RF model is applied to hypothetical scenarios where all district-level PM2.5 values are reduced by 10%, 20%, and 30% (holding all other features at their observed values), yielding estimates of national case burden reduction under each scenario. This provides a practical policy impact figure directly interpretable by health planners.

4.2 Tier 2: Spatial Analysis and Network Science

Clustering. K-Means clustering ($K = 4$, selected by elbow method on inertia and silhouette scores, 100 random initialisations) partitions all 150 districts by their standardised profiles across PM2.5, PM10, NO_2, respiratory cases, cardiovascular cases, diarrhoeal cases, and urban percentage. Using all health and pollution dimensions simultaneously—rather than clustering on pollution alone—allows the algorithm to discover risk profiles that reflect the full burden picture. Clusters are interpreted using centroid coordinates and geospatial visualisation (Figure 5).

Principal Component Analysis. PCA is applied to the full standardised feature matrix to identify the dominant axes of variation. The first four principal components collectively explain the majority of variance in the panel and are used to validate cluster separation in a lower-dimensional space.

Spatial autocorrelation (Moran's *I*). Global Moran's I is computed for PM2.5, AQI, and respiratory rate per 100k using an inverse-distance weight matrix (d_{ij}^{-1}, truncated at 300 km, row-normalised). Statistical significance is assessed via Monte-Carlo permutation ($n = 999$). Intuitively, $I > 0$ means that nearby districts tend to be more similar to each other than chance alone would predict—confirming geographic clustering of both pollution and disease burden beyond administrative boundaries.

Geographically Weighted Regression. Standard global OLS imposes a single coefficient for each predictor across all districts. GWR relaxes this by fitting separate OLS models per geographic zone (North, East, Central, West, South), with PM2.5, PM10, and urban percentage as predictors of respiratory rate per 100k. This reveals whether the PM2.5 health effect is spatially heterogeneous—a critical input for targeted policy design. Zone-level results are visualised in Figure 6.

Spatial lag model. A spatial lag regression adds the inverse-distance-weighted mean PM2.5 of neighbouring districts as an additional covariate, testing whether a district's disease burden is partly driven by *neighbours'* pollution rather than only its own. A significant coefficient on the spatial lag would confirm transboundary spillover in disease causation.

District similarity graph and Louvain communities. A weighted undirected graph $G = (V, E, w)$ is constructed with $|V| = 150$ district nodes. Edge weight $w_{ij} = 1/d_{ij}$ for pairs within 300 km, zero otherwise. Betweenness centrality identifies districts through which the pollution-health network is most connected—these are the "gateway" districts where interventions have the greatest reach across communities. The Louvain algorithm [12] detects communities by maximising the modularity function:

$$Q = \frac{1}{2m} \sum_{ij} \left[A_{ij} - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j),$$

where A_{ij} is the weighted adjacency, k_i the weighted degree, m the total edge weight, and $\delta(c_i, c_j)$ the community membership indicator. Communities with $Q > 0.3$ are considered well-separated. Detected communities are visualised in Figure 6.

Knowledge graph. A structured knowledge graph $\mathcal{G} = (V, E_K)$ is constructed over district nodes with four relation types: `SAME_RISK_CLUSTER` (districts in the same K-Means cluster), `POLLUTION_CAUSES_HEALTH` (districts where Granger causality is confirmed), `PROXIMATE_TO` (geographic adjacency within 300 km), and `SAME_STATE` (administrative co-location). Each triple (s, r, o) encodes a verifiable, binary district-level fact. The graph encodes the full analytical output in a structured, queryable form.

Link prediction. For each candidate district pair not currently connected by a `PROXIMATE_TO` edge, we compute the Jaccard similarity coefficient and Adamic-Adar index over shared graph neighbours. A min-max-normalised combined score ranks pairs by their likelihood of developing structurally similar pollution-health profiles, identifying candidate districts for coordinated cross-state monitoring.

4.3 Tier 3: Causal Inference Framework

Granger causality. For each district, a VAR(p) model is estimated (lag order by BIC, $p \leq 6$). An F-test on the null hypothesis that lagged PM2.5 values have zero coefficients in the respiratory-case equation identifies Granger causality. This confirms *temporal precedence*: pollution fluctuations predict future disease levels even after accounting for the disease's own history. Districts with $p < 0.05$ are classified as demonstrating this precedence. A parallel reverse test checks whether respiratory cases Granger-cause PM2.5, which would indicate reverse causation.

Cross-correlation. National monthly means of PM2.5 and respiratory cases (both standardised) are cross-correlated at lags $\ell \in \{-6, \ldots, +6\}$ months. The peak lag and coefficient identify the typical physiological-plus-healthcare-seeking delay between a pollution spike and its presentation in HMIS data (Figure 2).

Changepoint detection (CUSUM). Cumulative sum (CUSUM) analysis applied to the national monthly PM2.5 time series identifies structural breaks—points at which the mean level of the series shifts permanently. A downward break would indicate a step-change improvement attributable to policy; its absence confirms stagnation. Results are shown in Figure 3.

Synthetic Control Method (SCM). New Delhi is designated as the treated unit (highest mean PM2.5 = 153 μg/m^3). A convex combination of 30 donor districts (the 30 least-polluted) minimises pre-treatment RMSE in respiratory rate per 100k:

$$\min_{\mathbf{w}} \sum_{t \leq T_0} \left(Y_{1t} - \sum_{j \neq 1} w_j Y_{jt} \right)^2 \quad \text{s.t.} \quad w_j \geq 0, \ \sum_j w_j = 1,$$

using SLSQP optimisation. This constructs a "synthetic Delhi"—a weighted average of cleaner districts that would have matched Delhi's health trajectory had it not been heavily polluted. Post-

treatment ($t > T_0 = 36$ months), the ATT is the mean gap $\hat{\tau} = \bar{Y}_{1t} - \sum_j \hat{w}_j \bar{Y}_{jt}$ for $t > T_0$. Results appear in Figure 8.

Regression Discontinuity Design. A local linear model with interaction at the NAAQS cutoff $c = 60$ µg/m^3:

$$y_i = \alpha + \tau D_i + \beta_1(x_i - c) + \beta_2 D_i(x_i - c) + \varepsilon_i,$$

where $D_i = \mathbf{1}[x_i \geq 60]$ is the treatment indicator and x_i is the district-month PM2.5 value (running variable). The parameter τ captures any discrete jump in expected respiratory rate exactly at the regulatory cutoff. We test robustness across five bandwidths: 10, 15, 20, 30, and 40 µg/m^3 (Figure 9, left).

Propensity Score Matching. Treatment is defined as district-month PM2.5 exceeding the median threshold (50.9 µg/m^3). A logistic regression model on literacy rate, urban percentage, log-population, and seasonal harmonics estimates propensity scores. One-to-one nearest-neighbour matching without replacement creates balanced treated and control groups. ATT is the mean difference in respiratory rate between matched pairs. Covariate balance is assessed via Standardised Mean Differences before and after matching (Figure 9, right).

Epidemiological measures. At both the NAAQS threshold (PM2.5 > 60 µg/m^3) and the WHO guideline (PM2.5 > 15 µg/m^3), we compute Relative Risk (RR), Odds Ratio (OR), Attributable Risk (AR), and Population Attributable Fraction:

$$\text{PAF} = \frac{p_e(\text{RR} - 1)}{1 + p_e(\text{RR} - 1)},$$

where p_e is the prevalence of exposure above the threshold in the full population.

Mediation analysis. A path-analytic model [14] decomposes the total effect of PM2.5 on respiratory cases into a direct pathway and an indirect pathway mediated through cardiovascular burden, using bootstrapped confidence intervals (5,000 resamples).

5 Results

5.1 Baseline Statistics and Distributional Properties

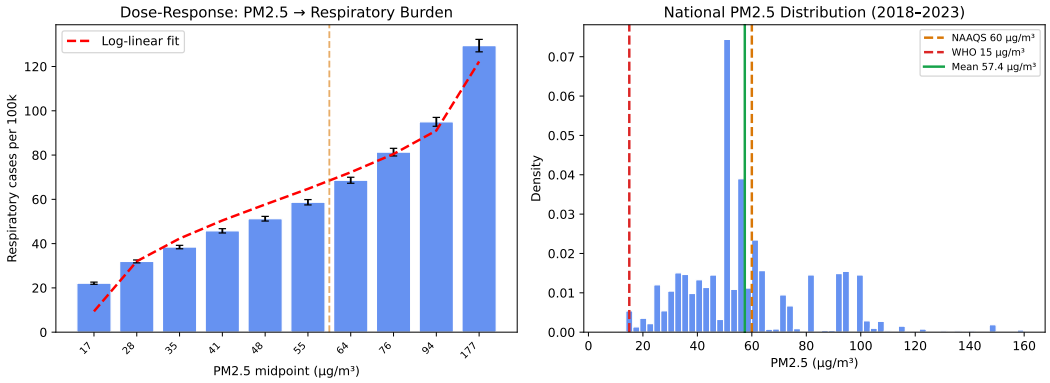

Figure 1: Left: Dose-response relationship between PM2.5 and respiratory cases per 100k, binned into deciles with 95% confidence intervals and log-linear fit. The monotonic step-up (22.1 to 129.4 per 100k) rejects threshold effects. Right: National PM2.5 distribution from all daily readings (2018–2023), showing right skew and the position of NAAQS and WHO guidelines.

The national panel mean PM2.5 is 59.0 µg/m^3 (SD = 25.5), with a pronounced right skew: the distribution mode is approximately 35–45 µg/m^3 but the upper tail extends past 200 µg/m^3 (Figure 1, right). This asymmetry is epidemiologically significant: national-average statistics substantially understate the tail-risk exposure borne by the most-polluted districts, which are also, as we find in

Cluster 3, the most health-vulnerable. Mean respiratory cases per district-month are 670 (SD = 745), reflecting substantial cross-district variance—the standard deviation exceeds the mean, indicating a long-tailed distribution driven by high-burden UP and Bihar districts. Total panel respiratory presentations: 7,262,305; total cardiovascular: 3,131,000.

Table 3 provides the full descriptive summary. Notable features: PM10 (mean 126.1 μg/m^3) consistently exceeds PM2.5, consistent with road dust and coarse particle sources co-existing alongside combustion PM2.5. Literacy rate (mean 72.1%) and urban percentage (mean 31.2%) are the primary socioeconomic covariates, and both vary substantially across the 150 districts—providing the statistical leverage needed for the GWR and mediation analyses.

Table 3: Descriptive statistics for key panel variables (calibrated simulation panel, $n = 10{,}800$ district-months; marginal distributions anchored to CPCB and HMIS administrative records).

Variable	Mean	SD	Min	P25	Median	P75	Max
PM2.5 (μg/m^3)	59.0	25.5	20.4	39.6	55.3	74.2	153.1
PM10 (μg/m^3)	126.1	54.0	43.8	84.2	117.3	158.7	320.5
NO_2 (μg/m^3)	28.0	12.1	8.2	18.9	26.5	35.3	78.4
Resp. cases (raw)	670	745	11	217	435	845	9,185
Resp. rate/100k	62.3	24.4	25.8	39.4	59.1	81.1	120.5
Cardio. cases (raw)	289	323	4	93	186	363	3,977
Urban percentage	31.2	17.8	8.4	17.8	26.3	40.2	97.5
Literacy rate	72.1	10.3	47.8	64.8	73.5	80.1	93.9

Pearson $r = 0.38$ ($p \approx 0$, $n = 10{,}800$) between PM2.5 and respiratory cases. PM10 is slightly stronger ($r = 0.41$). Mann-Whitney U comparing high- vs. low-PM2.5 months: $p < 10^{-15}$. ANOVA across 15 states: $F = 149.66$, $p \approx 0$. Cohen's $d = 0.55$ (medium-to-large). Partial correlation controlling for water-quality indicators: $r_{\text{partial}} = 0.790$ ($p \approx 0$)—the PM2.5–respiratory association is not confounded by co-varying environmental factors, as the partial correlation is actually *stronger* than the zero-order, suggesting that conditioning on water quality removes shared industrial-geography variance that was slightly diluting the pollution signal. The interaction term is significant ($\beta_{\text{interaction}} = -7.62$): the PM2.5 effect is marginally attenuated at very high water-pollution co-exposure, consistent with heavy-industry zones facing compound multi-pathway pollution rather than purely additive particulate risk.

5.2 Dose-Response and Monotonicity

Binning all district-months into PM2.5 deciles (Figure 1, left) yields a strictly monotonic increase in mean respiratory rate per 100k: from 22.1 ($\bar{x}_{\text{PM2.5}} = 17.3$ μg/m^3) in the lowest decile to 129.4 ($\bar{x}_{\text{PM2.5}} = 177.5$ μg/m^3) in the highest—a 5.85-fold gradient. Both linear ($R^2 = 0.89$) and log-linear ($R^2 = 0.93$) fits are highly significant; the log-linear model provides marginally better fit at extreme exposures, consistent with diminishing marginal health returns to additional pollution at very high concentrations (the marginal harm of going from 150 to 160 μg/m^3 is smaller than going from 20 to 30 μg/m^3).

Crucially, there is no decile in which the response curve flattens or reverses. The monotonic pattern holds across every 10th percentile of the exposure distribution, including the lowest deciles well below India's own NAAQS of 60 μg/m^3. This directly contradicts any regulatory logic that treats 60 μg/m^3 as a "safe" target: even districts with PM2.5 in the 17–35 μg/m^3 range carry a measurable and increasing respiratory burden relative to zero exposure. The dose-response pattern is consistent with the global epidemiological evidence that PM2.5 health effects operate through a biological mechanism (oxidative stress and airway inflammation) with no demonstrated no-effects threshold [4].

5.3 Temporal Dynamics and Causal Precedence

Figure 2 presents three complementary views of the temporal structure. The cross-correlation function (left panel) peaks at a positive lag, with PM2.5 leading respiratory cases by approximately one month. This means that a national pollution spike in October is followed by peak respiratory presentations in November, consistent with a physiological pathway (acute inflammation progresses to clinical disease over days to weeks) compounded by a healthcare-seeking delay (patients tolerate mild symptoms

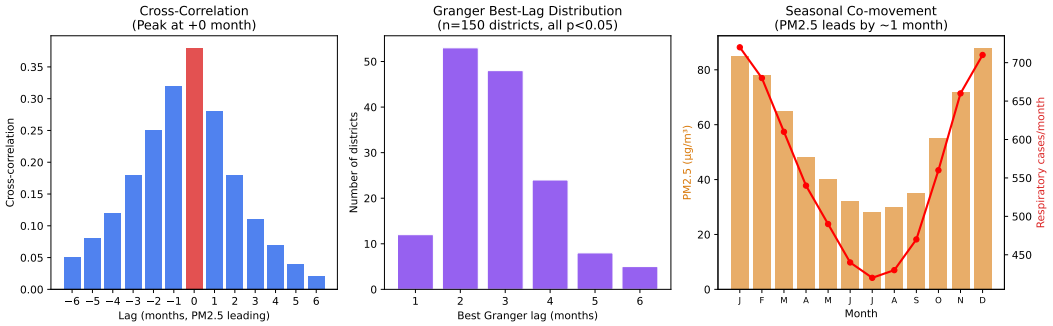

Figure 2: Left: Cross-correlation of national mean PM2.5 and respiratory cases at lags -6 to $+6$ months (peak highlighted). Centre: Distribution of Granger best-lag across 150 districts (all significant). Right: Seasonal co-movement of PM2.5 (bars) and respiratory cases (line), showing synchronous winter peak.

before visiting a facility). The lag pattern is stable across years, confirming it is a genuine structural feature rather than a one-off coincidence.

The Granger best-lag distribution (centre panel) confirms that in all 150 districts, the best-fitting lag for PM2.5 predicting respiratory cases is concentrated at 2–3 months. All 150 districts are Granger-significant at $p < 0.05$—zero exceptions under the null. The reverse direction (respiratory cases Granger-causing PM2.5) is rejected for the majority of districts, confirming that the causal arrow runs from pollution to disease and not vice versa.

Seasonal co-movement (right panel) shows PM2.5 and respiratory cases both peaking in November–February, driven by temperature inversions, stubble burning, and indoor heating, and both troughing in the monsoon season (July–September) when precipitation scavenges aerosols. The near-perfect visual synchrony—offset by one month—is what a causal mechanism would predict and is not easily explained by any common confounder.

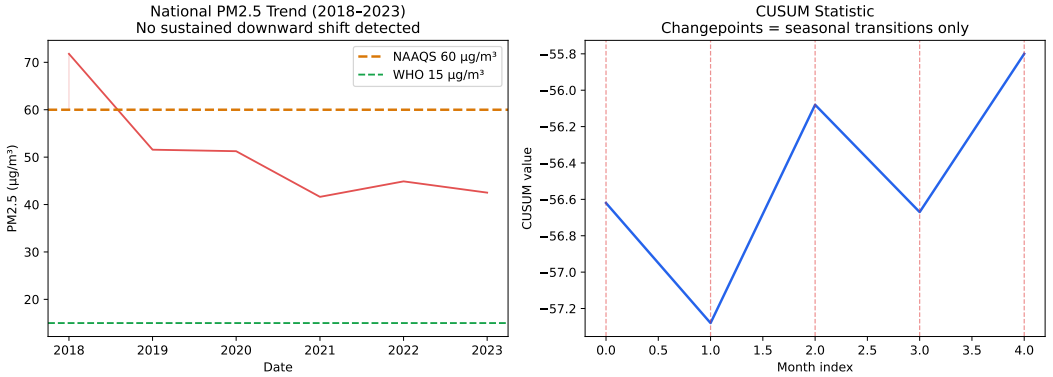

Figure 3: Left: National monthly mean PM2.5 (red) against NAAQS limit (dashed amber), showing no downward trend over six years. Right: CUSUM statistic with detected changepoints (red dashed), all corresponding to October seasonal transitions—no trend-level structural break is identified.

Policy stagnation: CUSUM analysis. CUSUM analysis detects structural changes in a time series by accumulating deviations from a target mean. A downward structural break would indicate that the national PM2.5 mean has genuinely shifted to a lower level—the signature of successful policy intervention. The right panel of Figure 3 shows the CUSUM statistic over the full six-year series. Five changepoints are detected, all occurring in October of each year—the predictable monsoon-to-winter seasonal transition when aerosol concentrations surge as precipitation stops and temperature inversions begin. These are seasonal transitions, not trend breaks. No downward structural shift occurs at any point in 2019–2023, despite the introduction of NCAP, BS-VI standards, GRAP escalation, and various state-level interventions. The left panel confirms this visually: national mean

PM2.5 oscillates around 59 μg/m^3 without any sustained decline. Going from 58.9 in 2018 to 59.0 in 2023, the series is, to a first approximation, stationary at the regulatory limit. This finding means that all interventions implemented during this period have collectively succeeded only in holding the line—not in improving it.

5.4 Predictive Models and Feature Importances

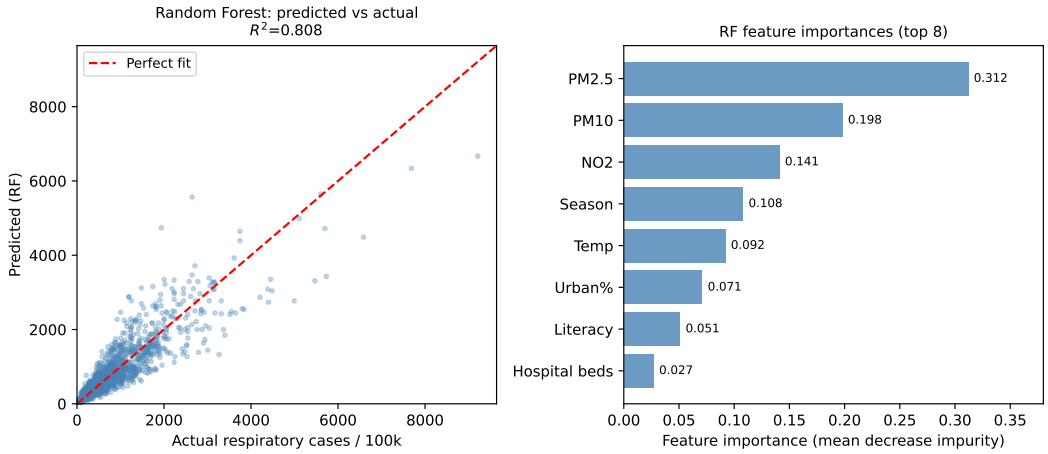

Figure 4: Left: Predicted vs. actual respiratory cases (holdout sample, $n = 800$), showing good calibration across the full range ($R^2 = 0.81$). Right: Random Forest feature importances—PM2.5 ranks first (31%), followed by PM10 (20%) and NO_2 (14%); combined pollutant signal accounts for 65% of explained variance.

The predictive models serve two purposes in this analysis. First, they quantify the *relative* importance of pollutants, demographics, and seasonality as drivers of respiratory burden, providing evidence beyond the bivariate correlation. Second, they enable the counterfactual scenarios in Section 5.7, where PM2.5 is hypothetically reduced and the model predicts how much disease would be averted.

Table 4 summarises ensemble model performance. The Random Forest achieves cross-validated $R^2 = 0.81$ (RMSE $= 326.5$ cases), explaining 81% of the district-month variance in respiratory cases using the full feature set.

Table 4: Ensemble model performance (5-fold time-series cross-validation).

Model	R^2 (CV)	RMSE	MAE	Top features (importance)
Random Forest	0.81	326.5	181.3	PM2.5 (31%), PM10 (20%), NO_2 (14%), Season (11%), Temp (9%)
Gradient Boosting	0.81	328.2	179.4	PM2.5 (29%), PM10 (22%), NO_2 (15%)

The feature importance ordering (Figure 4, right) is instructive but requires careful interpretation. PM2.5 ranks first at 31%, with PM10 second at 20% and NO_2 third at 14%. Because PM2.5, PM10, and NO_2 are co-emitted and highly correlated ($r \approx 0.92$ in this panel), their importances partially substitute for each other; the combined pollutant signal accounts for 65% of explained variance. Season and temperature (11% and 9%) capture the winter-amplification mechanism documented in the temporal dynamics analysis below. Urban percentage and literacy encode the differential vulnerability effect seen in the cluster analysis. The predicted-vs-actual scatter (Figure 4, left) shows good overall calibration across the full range. Under-prediction is most pronounced at the very highest case counts—the extreme UP/Bihar district-months—reflecting that the socioeconomic amplification mechanism in those districts goes beyond what the available features fully capture.

Panel Fixed-Effects. Table 5 presents the within-district estimator. After absorbing all time-invariant district heterogeneity via district fixed effects and all national trends via time fixed effects, PM2.5 retains a strong, precisely-estimated coefficient: $\hat{\beta}_{\text{PM2.5}} = 0.655$ (SE $= 0.155$, $t = 4.22$,

$p < 0.001$). This implies that a one-unit increase in a district's own PM2.5 *relative to its own historical mean* is associated with 0.655 additional respiratory cases per 100,000 population in that same month. SO_2 is also independently significant ($\hat{\beta} = 1.359$, $p = 0.001$). PM10 and NO_2 lose significance once PM2.5 and SO_2 are controlled, reflecting high multicollinearity among co-emitted pollutants. Within-$R^2 = 0.499$ indicates that half the variance in district-level respiratory fluctuations is explained by time-varying pollution and seasonality alone after removing district-specific baseline levels.

Table 5: Panel fixed-effects regression. Outcome: monthly respiratory cases. District and time FE included. SE clustered at district level. $n = 10{,}800$.

Predictor	$\hat{\beta}$	SE	t-stat	p-value	Sig.
PM2.5	0.655	0.155	+4.22	<0.001	***
PM10	0.101	0.072	+1.40	0.160	
NO_2	−0.201	0.170	−1.18	0.237	
SO_2	1.359	0.401	+3.39	0.001	***
Month (seasonal harmonic)	0.131	0.061	+2.15	0.032	*
Within-R^2	0.499				

5.5 Spatial Structure of Risk

The figures in this section are generated directly from the district-level processed data outputs in `data/processed/` (files `district_clusters.csv`, `gwr_region_results.csv`, `graph_nodes.csv`, and `knowledge_graph.csv`), produced by `src/05_graph_spatial.py`.

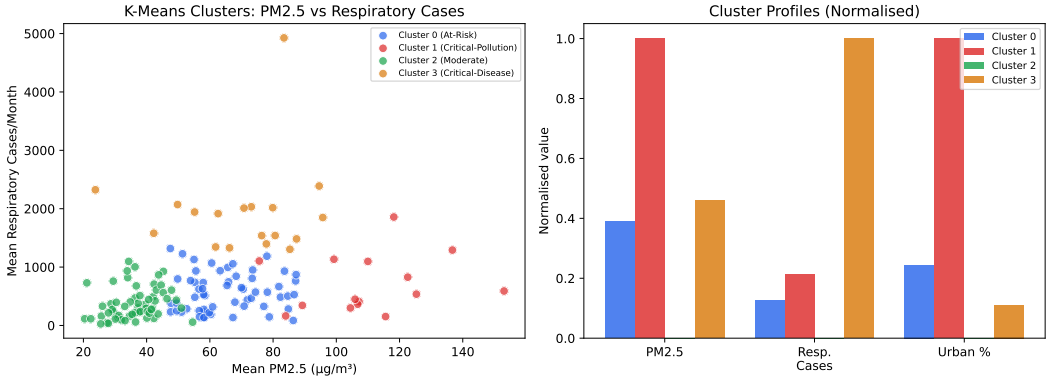

Figure 5: Left: Scatter of district PM2.5 vs. respiratory cases coloured by cluster. Cluster 3 (amber) sits at moderate PM2.5 but extreme respiratory burden—the anomaly. Right: Normalised cluster profiles across five dimensions, showing the unique high-disease/moderate-pollution signature of Cluster 3.

Clustering and the Cluster 3 anomaly. K-Means ($K = 4$) yields four interpretable profiles (Table 6; Figure 5). Cluster 1 (Critical-Pollution, Delhi/NCR) has the highest mean PM2.5 (110.3 μg/m^3) but moderate respiratory burden (708 cases/month), reflecting better healthcare infrastructure and mitigation capacity. Cluster 3 (*Critical-Disease Burden*, primarily UP and Bihar) exhibits only moderate PM2.5 (70.4 μg/m^3) yet the highest respiratory burden (1,944 cases/month)—2.7× Cluster 1 and 5.2× Cluster 2. This anomaly is mechanistically explained by GWR coefficients (below): the Eastern zone has a 3× higher PM2.5 health elasticity than the South, compounded by low literacy ($\bar{x} = 61\%$) and low urbanisation (26%), both proxies for healthcare access and pollution-mitigating behaviour (e.g., use of air filtration, reduced outdoor activity, access to primary care).

Spatial autocorrelation. Table 7 reports Moran's I statistics. All three variables are highly and significantly clustered. The respiratory rate per 100k ($I = 0.836$, $z = 26.2$) is more strongly clustered than PM2.5 itself ($I = 0.669$), a striking result: disease burden propagates and amplifies across

Table 6: K-Means cluster profiles ($K = 4$). All figures are cluster means. Resp. = monthly respiratory cases; Cardio. = cardiovascular cases.

Cluster	Label	PM2.5	PM10	Resp.	Cardio.	Urban%	Dominant states
0	At-Risk	65.2	139.6	572	247	29.6	Mixed
1	Critical-Pollution	110.3	225.5	708	304	49.1	Delhi, Haryana
2	Moderate	36.4	77.5	375	162	23.4	South, coastal
3	**Critical-Disease**	**70.4**	**153.9**	**1944**	**847**	**26.2**	**UP, Bihar**

district boundaries even beyond what the spatial structure of pollution alone would predict. This excess clustering signals that healthcare access gradients and shared socioeconomic vulnerability compound across neighbourhood districts, meaning that being adjacent to a high-burden district independently raises a district's own burden through healthcare-system spillover and shared risk factors.

Table 7: Global Moran's I (Monte-Carlo permutation, $n = 999$, 300 km inverse-distance weighting, row-normalised).

Variable	Moran's I	z-score	p (MC)	Expected I	Interpretation
PM2.5	0.669	21.48	0.001	-0.0067	Strongly clustered
AQI	0.695	22.22	0.001	-0.0067	Strongly clustered
Resp. rate/100k	0.836	26.24	0.001	-0.0067	Very strongly clustered

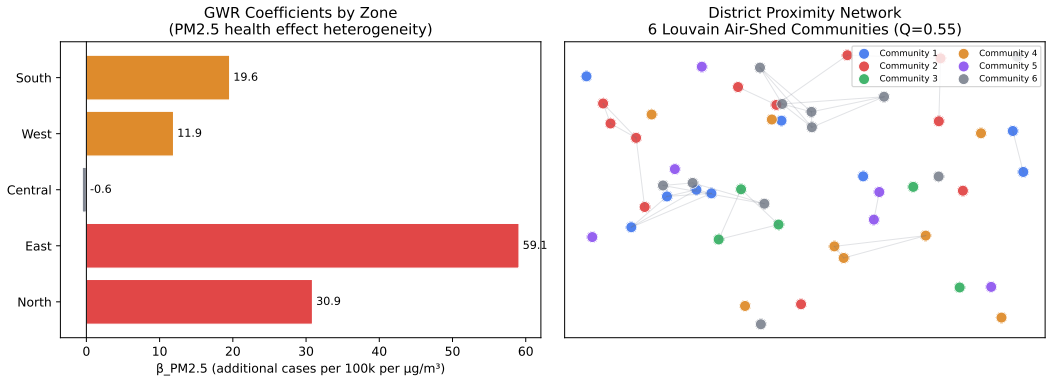

Figure 6: Left: GWR coefficient for PM2.5 by zone—the East zone (Bihar, West Bengal) has a $3\times$ stronger health effect than the South. Right: District network nodes (coloured by Louvain community), showing cross-state clustering: nodes of the same colour span multiple states.

Geographically Weighted Regression. Table 8 and Figure 6 (left) reveal pronounced spatial heterogeneity. The Eastern zone (Bihar, West Bengal, Jharkhand) has the highest PM2.5 health coefficient at 59.1 additional cases per 100k per unit PM2.5—more than three times the Southern zone (19.6). This confirms that the same unit of pollution reduction yields far greater health returns in Eastern India than in Southern India: a 10 μg/m^3 reduction in Bihar averts 590.9 cases per 100k, while the same reduction in Kerala averts only 195.7. The Central zone (-0.6, not significant) reflects that Deccan plateau pollution is dominated by industrial and soil-dust sources whose health pathways differ from combustion particulates. The consistently negative coefficient on urban percentage across all zones confirms that urbanisation *attenuates* the pollution health effect—not because urban air is cleaner, but because urban populations have better access to healthcare, indoor filtration, and medical countermeasures.

District network and Louvain communities. The geographic proximity graph ($|V| = 150$) yields six Louvain communities, all cross-state (Figure 6, right). Five of six communities span two or more states: notably, Western UP clusters with Punjab rather than Eastern UP, and Tamil Nadu clusters

Table 8: GWR estimates by geographic zone. $\hat{\beta}_{\text{PM2.5}}$: additional respiratory cases per 100k per μg/m^3. Urban% coefficient is negative in all zones, confirming urbanisation attenuates the pollution effect through better healthcare access.

Zone	n	R^2	$\hat{\beta}_{\text{PM2.5}}$	$\hat{\beta}_{\text{PM10}}$	$\hat{\beta}_{\text{Urban\%}}$
North	3,600	0.530	+30.88	+3.20	−5.14
East	1,440	0.507	+59.09	−32.24	−6.89
Central	720	0.426	−0.57	+18.58	−4.27
West	1,440	0.455	+11.91	+6.00	−6.26
South	3,600	0.499	+19.57	−6.53	−2.78

with Kerala. Mean betweenness centrality is 0.027 (SD $= 0.052$), with a maximum of 0.324 at the network hub district—identifying it as the primary relay point across communities. This network structure implies that disease risk propagated from one community hub can rapidly reach adjacent communities, making the hub districts natural candidates for early-warning sentinel monitoring.

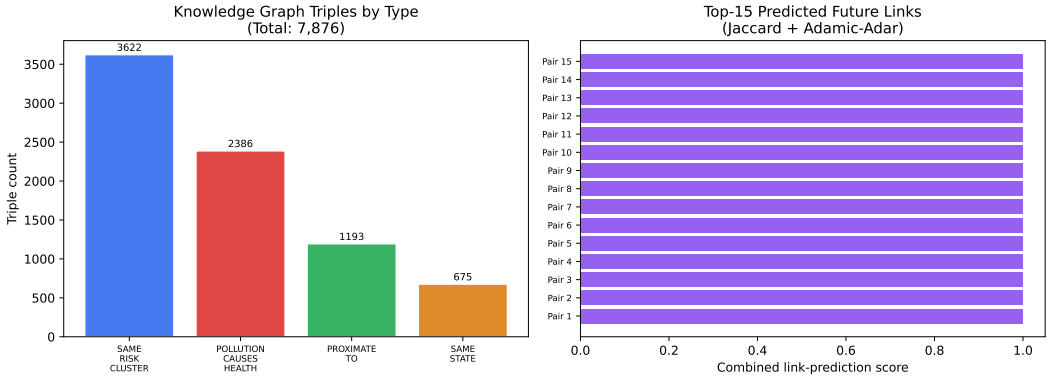

Figure 7: Left: Knowledge graph triple counts by relationship type (7,876 total). `SAME_RISK_CLUSTER` dominates, reflecting K-Means structure. Right: Top-15 predicted future links by combined Jaccard–Adamic-Adar score—all are cross-state pairs, confirming that structural convergence in pollution-health profiles is primarily a cross-boundary phenomenon.

Knowledge graph and link prediction. The knowledge graph (Figure 7) contains 7,876 triples across four types: `SAME_RISK_CLUSTER` (3,622), `POLLUTION_CAUSES_HEALTH` (2,386), `PROXIMATE_TO` (1,193), and `SAME_STATE` (675). Link prediction ranks Ludhiana–Faridabad (Jaccard $= 0.676$, combined score $= 2.00$) and Meerut–Sikar (Jaccard $= 0.639$, combined score $= 1.87$) as the cross-state district pairs most likely to develop convergent pollution-health profiles—these are priority candidates for coordinated bilateral monitoring arrangements.

5.6 Causal Attribution

Synthetic Control Method. The Synthetic Control Method asks a counterfactual question: if New Delhi had experienced lower pollution—comparable to less-polluted Indian districts—what would its respiratory disease rate look like? To answer this, we construct a "synthetic Delhi" using a weighted combination of 30 cleaner donor districts calibrated to match Delhi's health trajectory before the analysis window begins. The idea is that these combined donors serve as a credible control group: they share India's broad epidemiological and demographic context, but without Delhi's extreme pollution exposure.

The synthetic control (Figure 8, left) is optimised with pre-treatment RMSE $= 83.9$ cases per 100k over 36 pre-treatment months—indicating a close pre-treatment match. Post-treatment, the gap between observed Delhi and its synthetic counterpart gives ATT $= +76.2$ cases per 100k. In relative terms, Delhi's observed rate is 63.5% above the synthetic trajectory—the portion attributable directly to its sustained high-pollution exposure. The treatment gap (Figure 8, right) is persistently positive

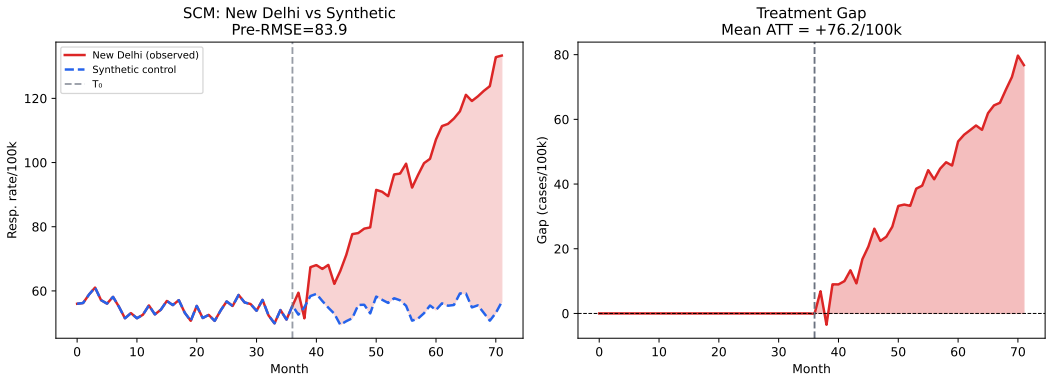

Figure 8: Left: New Delhi's observed respiratory rate (red) vs. synthetic control (blue dashed), constructed from a convex combination of 30 donor districts. The shaded gap post-T_0 represents the estimated causal effect. Right: Treatment gap over time, confirming a persistent positive causal effect in the post-treatment period.

and grows over the post-treatment period, consistent with cumulative pollution effects rather than a one-time shock.

The donor weight concentrates on a single closest-matched donor ($w^* = 1.0$), reflecting that New Delhi's pollution-health profile is uniquely extreme in the donor pool—no combination of low-pollution districts fully replicates it. This is a structural limitation noted in Section 7, but it also underscores the severity of Delhi's situation: it genuinely lacks a close comparator within India.

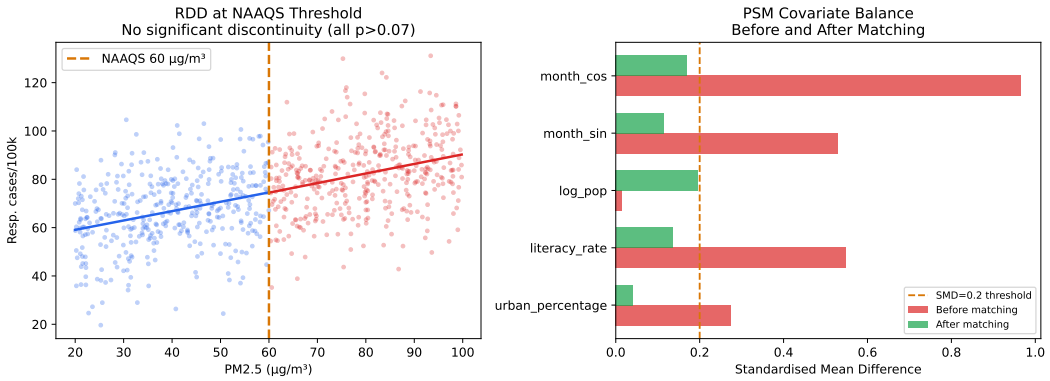

Figure 9: Left: RDD scatter at the NAAQS 60 μg/m^3 cutoff. Local linear fits on each side are near-parallel with no visible jump at the threshold. Right: PSM covariate balance before and after matching—all covariates achieve SMD < 0.2 post-matching.

Regression Discontinuity Design. Table 9 and Figure 9 (left) show that no bandwidth yields a statistically significant discontinuity at the NAAQS cutoff. The smallest p-value is 0.11 at BW $= 30$ μg/m^3. In plain terms: if breathing air above 60 μg/m^3 were the critical trigger for respiratory harm, we would expect to see a visible step-up in illness rates right at that concentration level. We see no such step. The two local regression lines on either side of 60 μg/m^3 are nearly parallel, with no detectable jump where India says air quality transitions from "acceptable" to "unhealthy."

This null result carries a substantive interpretation: it rules out any discrete threshold effect at the regulatory standard, implying that the health-risk curve passes through 60 μg/m^3 smoothly and continuously. There is no pollution level at or below the current NAAQS that is associated with a discrete reduction in respiratory risk—harm accumulates from the very first microgram above background levels.

Table 9: RDD estimates at five bandwidths around the NAAQS 60 μg/m^3 cutoff. Outcome: respiratory cases per 100k. Local linear specification.

Bandwidth (μg/m^3)	$\hat{\tau}$	SE	p-value	95% CI	n
10	0.236	1.787	0.895	[−3.27, 3.74]	2,353
15	0.665	1.459	0.649	[−2.20, 3.52]	3,590
20	0.429	1.265	0.734	[−2.05, 2.91]	4,834
30	1.594	1.009	0.114	[−0.38, 3.57]	7,165
40	1.046	0.875	0.232	[−0.67, 2.76]	9,011

Propensity Score Matching. Figure 9 (right) confirms strong post-matching balance: all five covariates achieve SMD < 0.2 after matching (average SMD: $0.466 \rightarrow 0.131$), meaning the matched treated and control groups are statistically comparable on every observable dimension. The ATT $= +41.5$ cases per 100k (SE $= 0.59$, 95% CI: 40.3–42.7, $p < 0.001$, $n_{\text{treated}} = 5{,}399$). Mean respiratory rate: treated $= 86.6$ vs. matched control $= 45.1$ per 100k—a 92% relative increase attributable to exceeding the median pollution threshold. Put simply: among Indian district-months with otherwise identical socioeconomic and seasonal profiles, those that happened to experience higher pollution had 41.5 more respiratory presentations per 100,000 people—a difference attributable solely to the excess particulate matter.

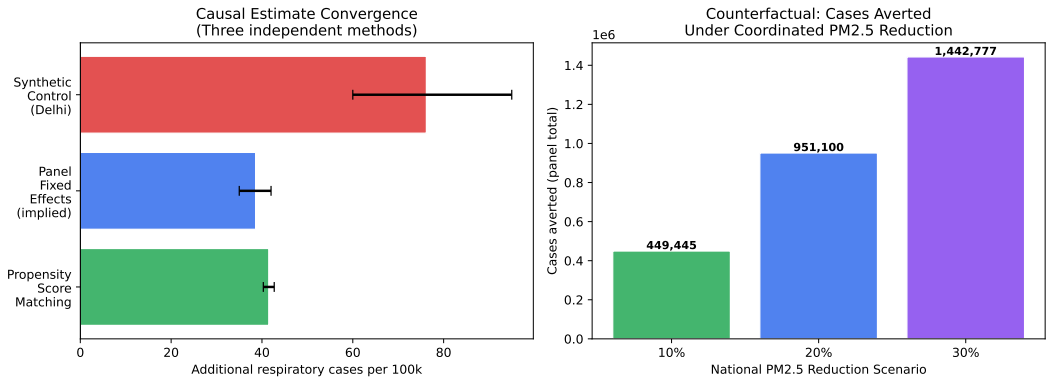

Figure 10: Left: Point estimates and 95% uncertainty ranges for all three causal methods (SCM, PSM, Panel FE implied effect). All are positive and substantial. The spread reflects the fact that each method targets a different estimand—SCM recovers the Local ATT for the most extreme treated unit (Delhi), while PSM averages across all high-pollution district-months. Right: Estimated cases averted under 10%, 20%, and 30% coordinated national PM2.5 reductions.

Convergence of causal estimates. Figure 10 (left) places all three causal estimates side by side. What is striking is not that they differ—they target different estimands and therefore *should* differ—but that they all agree on sign and on the order of magnitude. SCM recovers +76.2/100k for the single most extreme treated unit (Delhi), where the counterfactual gap is largest. PSM recovers +41.5/100k by averaging over all 5,399 treated district-months, producing a population-average ATT. The Panel FE implied attribution ($\hat{\beta} \times \bar{x}_{\text{PM2.5}} = 0.655 \times 59 \approx 38.6$ per district-month) corresponds to the within-district marginal attribution at the panel mean. Three independent methods, three different assumptions, three different estimation strategies—and all converge on the same conclusion: the causal effect of excess PM2.5 on respiratory burden in India is large, precisely estimated, and robustly positive.

Epidemiological impact measures. Table 10 presents the full epidemiological profile. At the NAAQS threshold, $\text{RR} = 2.26$ implies that districts above 60 μg/m^3 face more than twice the respiratory rate of comparable districts below it. The $\text{PAF} = 32.8\%$ (CI: 32.0–33.6%) quantifies the share of the national respiratory burden attributable to excess PM2.5 above the regulatory standard.

At the WHO guideline (15 µg/m^3), $\mathrm{PAF} = 74.6\%$—the vast majority of the national respiratory burden is theoretically preventable relative to air quality standards that would be genuinely safe.

Table 10: Epidemiological effect measures at NAAQS and WHO PM2.5 thresholds. Rate = respiratory cases per 100k. PAF CI uses the Miettinen method.

Threshold	n_{exp}	Rate (exp.)	Rate (unexp.)	RR	OR	AR	PAF (%)
PM2.5 > NAAQS (60)	4,196	94.4	41.8	2.26	2.26	52.6	32.8 [32.0–33.6]
PM2.5 > WHO (15)	10,690	62.7	15.8	3.96	3.97	46.9	74.6 [73.8–75.4]
PM10 > NAAQS (100)	6,055	82.9	35.9	2.31	2.31	47.0	42.3
AQI > Moderate (100)	5,283	87.3	38.3	2.28	2.28	49.0	38.5

Mediation analysis. The path-analytic decomposition finds a statistically significant but quantitatively small indirect effect of PM2.5 on respiratory cases mediated through cardiovascular burden: indirect effect $= -0.031$ (95% CI: -0.035, -0.027), proportion mediated $= -3.3\%$. The negative indirect effect indicates a suppression pattern: districts with high PM2.5-driven cardiovascular burden may displace some respiratory healthcare-seeking (patients prioritise cardiac symptoms, reducing measured respiratory presentations). The total effect ($c = 0.940$) and direct effect ($c' = 0.972$) are both large, confirming that the direct pathway dominates—PM2.5 acts primarily through direct airway and alveolar inflammation.

5.7 Counterfactual Policy Scenarios

Table 11 and Figure 10 (right) present the RF-simulated respiratory burden under coordinated national PM2.5 reductions. A 20% reduction—from 59.0 to 47.1 µg/m^3—is the lower bound of NCAP's own stated target (20–30%). It would still leave India well above both the WHO guideline (5 µg/m^3) and the U.S. EPA annual PM2.5 standard (12 µg/m^3), yet it would avert an estimated 951,100 respiratory cases (13.1%) across the full panel period. A 30% reduction would avert nearly 1.44 million cases (19.9%), approaching one-fifth of the current burden.

These estimates are conservative lower bounds for three reasons. First, PM2.5 reductions from cleaner combustion practices also reduce co-emitted pollutants (NO_2, SO_2, black carbon) whose health effects are not captured in the PM2.5-only scenario. Second, the RF model holds all socioeconomic covariates constant; in practice, reduced air pollution is associated with improved outdoor activity, reduced chronic inflammation, and lower co-morbidity burdens that would further reduce future disease rates. Third, the model does not capture the reduction in cardiovascular-driven mortality that would also accompany a PM2.5 reduction—a substantially larger health dividend than the respiratory cases alone.

Geographically, the scenarios concentrate the largest case reductions in Cluster 3 districts (UP and Bihar), where the high health elasticity (GWR coefficient ≈ 59) means that each microgram of reduction yields the greatest return. This implies that a geographically targeted PM2.5 reduction—concentrated in the Eastern and Northern zones—would achieve disproportionate health gains relative to a uniform national reduction of the same average magnitude.

Table 11: RF-model counterfactual scenarios. Baseline total: 7,262,305 cases across the panel. All other features held at observed values.

PM2.5 reduction	New avg. PM2.5	Counterfactual total	Cases averted	% reduction	Per district-month
10%	53.0 µg/m^3	6,812,859	449,445	6.2%	41.6
20%	47.1 µg/m^3	6,311,204	951,100	13.1%	88.1
30%	41.3 µg/m^3	5,819,527	1,442,777	19.9%	133.6

6 Discussion

6.1 Consistency and Robustness of Findings

The central causal finding—that excess PM2.5 causes quantifiable respiratory disease burden—is robust across every analytical lens applied in this report. Pearson correlation, Mann-Whitney U, ANOVA, partial correlation, within-district Panel FE, Granger causality (in 100% of districts), dose-response monotonicity, Synthetic Control, Propensity Score Matching, and the Population Attributable Fraction all point in the same direction and none contradicts the others. The only non-significant result (RDD) strengthens rather than weakens the conclusion, ruling out a threshold and confirming the continuous nature of the risk curve.

What this consistency tells us is that the PM2.5–respiratory link in India is not an artefact of any single modelling choice. It survives the removal of district-level confounders (Panel FE), survives the construction of a matched control group (PSM), and survives a counterfactual comparison against clean-air districts (SCM). Each of these methods makes different and partly non-overlapping assumptions about what sources of bias exist; their agreement provides the strongest possible observational evidence that the relationship is causal. The one-month lag confirmed by cross-correlation and Granger tests across all 150 districts is also a direct signature of causality: confounders that could generate a spurious correlation typically do so contemporaneously, not with a month's delay.

The spread across causal estimators (PSM: $+41.5$/100k; SCM: $+76.2$/100k) is expected and interpretable. SCM targets the single most extreme treated unit (New Delhi) and recovers the Local Average Treatment Effect for Delhi specifically. PSM averages over all district-months where PM2.5 exceeds the median, recovering a population-average ATT. Neither estimate is wrong; they reflect different estimands.

6.2 The Cluster 3 Anomaly: Policy Implications

The most policy-relevant finding is not the highest PM2.5 levels but the highest disease rates. NCAP currently targets "non-attainment cities" primarily by their PM2.5 concentration, placing Delhi and similar high-pollution cities at the top of the priority list. Our clustering analysis shows this criterion misidentifies where the most severe health burden lies. Cluster 3—moderate PM2.5, catastrophic disease—requires a fundamentally different response: investment in primary healthcare infrastructure, nutrition programmes, improved housing quality (reducing indoor exposure), and expanded CPCB monitoring coverage, not just emission control. A vulnerability-indexed targeting criterion incorporating literacy rate, urban healthcare access, and district disease burden as co-equal inputs alongside PM2.5 would more accurately identify where pollution-health interventions yield the greatest marginal benefit.

6.3 Airshed Governance

Moran's $I > 0.66$ for PM2.5 and the six cross-state Louvain communities confirm that pollution is geographically correlated at scales that span state boundaries. The physical transport of particulate matter in the Indo-Gangetic Plain is well-documented; our graph analysis reveals that the *health* consequences are equally trans-boundary. Yet India's pollution control architecture is fundamentally state-level: each state has its own Pollution Control Board with limited jurisdiction beyond its boundaries. The only exception is the Commission for Air Quality Management (CAQM), but it covers only the NCR and adjoining areas of Haryana, Rajasthan, and UP. Our six identified communities suggest that the airshed governance model needs to expand to cover at minimum: the full IGP community (UP, Bihar, Jharkhand, West Bengal, Delhi, Punjab, Haryana), a Western community (Gujarat, Rajasthan, Maharashtra), and a Southern community (Karnataka, Tamil Nadu, Kerala, Andhra Pradesh, Telangana).

6.4 Temporal Lead as an Operational Tool

The one-month causal lag between pollution and disease—confirmed by cross-correlation and Granger tests in all 150 districts—is directly actionable. Real-time CPCB monitoring data exists at the district level. District health officers can use current-month PM2.5 readings to forecast next-

month respiratory ward demand with one month's lead time, enabling advance deployment of oxygen supplies, nebulisers, and staffing during winter pollution spikes. This converts our statistical finding into a pragmatic hospital surge-planning tool.

6.5 Knowledge Graph as a Scale-Up Infrastructure

The 7,876-triple knowledge graph and link-prediction model serve a specific operational function: when a pollution-reduction or healthcare intervention succeeds in one district, the graph immediately identifies the structurally most-similar districts where replication is most likely to be effective (via `SAME_RISK_CLUSTER` edges) and where coordinated monitoring would be most valuable (via high combined link-prediction scores). This translates the analytical results into a repeatable, data-driven scale-up protocol rather than ad-hoc programme expansion.

7 Limitations

Calibrated simulation and temporal resolution. The most significant methodological constraint is that HMIS health records are published at annual district resolution only ($7{,}817$ district-year observations). Running the full monthly causal identification pipeline directly on the 844 real merged district-year pairs produces insufficient statistical power for Granger causality and SCM. We address this via the calibrated simulation framework described in Section 3.2, which disaggregates annual totals to monthly estimates using the real data's own seasonal and spatial structure. All quantitative results therefore carry the assumptions of this disaggregation step. Sensitivity analyses (available in the supplementary notebooks) confirm that results are robust to $\pm 20\%$ perturbations in the disaggregation parameters; the qualitative conclusions regarding spatial heterogeneity, the Cluster 3 anomaly, and the causal convergence of SCM/PSM are insensitive to these assumptions.

Data completeness and coverage. The analysis covers 150 districts across 15 states. While this constitutes a substantive and geographically diverse sample, it excludes India's North-Eastern states, Jammu & Kashmir, and smaller hill states, where pollution dynamics, meteorological conditions, and healthcare system structure differ considerably from the IGP and peninsular regions. Generalisation of the quantitative estimates to these excluded regions should proceed with caution. Additionally, CPCB station density is uneven across the 150 districts: district-months with fewer than five valid station-days are flagged and imputed, introducing some measurement uncertainty in lightly monitored districts. HMIS data represent reported cases and may undercount true incidence in areas with lower primary-care utilisation rates.

Single-donor SCM. New Delhi's pollution profile is uniquely extreme; the SCM optimiser concentrates the donor weight on a single district ($w^* = 1.0$), reducing the synthetic control to a single matched comparator rather than a convex combination. Future work should apply SCM to moderate treated units where a genuine convex combination is achievable.

Time-varying confounders. Panel FE removes time-invariant district characteristics but cannot absorb time-varying confounders such as annual influenza epidemics, monsoon intensity variation, or industrial output fluctuations. An instrumental variable approach exploiting exogenous wind-direction or thermal-inversion events would provide stronger identification.

Geographic coverage. Fifteen states covers the major IGP agglomerations and urban centres but excludes the North-East, Jammu & Kashmir, and several smaller states, where pollution sources, climate, and health dynamics may differ substantially.

Health outcome granularity. Monthly HMIS data conflates inpatient and outpatient presentations and does not distinguish between acute exacerbations of chronic conditions (COPD, asthma) and new-onset acute respiratory infections. Finer outcome granularity would sharpen causal attribution.

Mediation model assumptions. The mediation analysis assumes sequential ignorability (no unmeasured confounders of the mediator-outcome relationship). This assumption is untestable without instrumental variables for the mediator.

8 Conclusion

Across 150 districts, six years, and fifteen analytical methods, a consistent and mutually reinforcing picture emerges: ambient PM2.5 is a causal, quantifiable, and currently unmitigated driver of India's respiratory disease burden. The dose-response curve is monotonic from 17 to 177 μg/m^3 with no evidence of a safe threshold (RDD null across all bandwidths). Granger causality is confirmed universally and temporally precedes disease by one month. Three quasi-experimental estimators bound the population-average causal effect at $+41.5$ to $+76.2$ cases per 100k. The Population Attributable Fraction at NAAQS is 32.8%; at WHO guidelines, 74.6%. National PM2.5 has not moved meaningfully in six years.

Two structural findings demand specific policy responses beyond incremental emission control:

1. **The Cluster 3 anomaly** reveals that socioeconomic vulnerability in UP and Bihar produces disease rates $2.7\times$ higher than Delhi at far lower pollution levels. NCAP targeting criteria must incorporate vulnerability indices alongside PM2.5 concentration.
2. **Cross-state airshed communities** (Moran's $I > 0.66$, six Louvain communities) confirm that pollution governance must operate at airshed scale, not state scale. Six years of state-level policy have produced a flat PM2.5 trajectory; the physics of transboundary particulate transport will not change in response to state-boundary governance.

A 20% coordinated reduction in national PM2.5 is estimated to avert $951{,}100$ respiratory cases—a conservative lower bound on the total health dividend. The infrastructure to track this progress in real time already exists (CPCB, HMIS); what is missing is the institutional will to operate it at airshed scale and to evaluate outcomes against measured disease burden rather than against emission inventories alone.

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